DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM
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ALGEBRA (Ph.D. Version)

Instructions to the student

a. Answer all six questions; each will be assigned a grade from 0 to 10.

b. Use a different booklet for each question. Write the problem number and your **code number (not** your name) on the outside of the booklet.

c. Keep scratch work on separate pages in the same booklet.

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1. (a) Let G be a group. Suppose that every cyclic subgroup of G is normal in G. Show that every subgroup of G is normal in G.

(b) Let $H = \{\pm 1, \pm i, \pm j, \pm k\}$, with $ij = k = -ji, i^2 = j^2 = k^2 = -1$, be the quaternion group with 8 elements. Let A be an abelian group of odd order and let $G = H \times A$. Show that every subgroup of G is normal in G.

2. Let A and B be linear transformations from $\mathbb{C}^n$ to $\mathbb{C}^n$. Let N_A and N_B be the nullspaces of A and B.

(a) Show that if $N_A \cap N_B \neq 0$, then $\det(xA - B) = 0$ for all complex numbers x.

(b) Assume A and B are diagonalizable and $AB = BA$. Suppose that $\det(xA - B) = 0$ for all complex numbers x. Show that $N_A \cap N_B \neq 0$. (You may use without proof the well-known fact that there is a basis for which both A and B are diagonal.)

3. Let R be a commutative Noetherian ring with 1 (one definition of Noetherian is that every non-empty collection of ideals of R contains an ideal that is maximal for that set). Let M be a non-zero R-module. If $m \in M$, define the annihilator of m to be $\mathrm{Ann}(m) = \{r \in R \,|\, rm = 0\}$.

(a) Suppose I is an ideal of R that is maximal among ideals that are annihilators of nonzero elements (that is, $I = \mathrm{Ann}(m)$ for some $0 \neq m \in M$, and if $I \subseteq \mathrm{Ann}(x)$ for some $0 \neq x \in M$, then $I = \mathrm{Ann}(x)$). Show that I is a prime ideal.

(b) Let $r \in R$ and $0 \neq m \in M$ be such that $rm = 0$. Show that there exists a prime ideal P of R and a nonzero element $x \in M$ such that $r \in P$ and $P = \mathrm{Ann}(x)$.

4. Let R be a commutative ring with 1. Recall that an R-module P is projective if whenever $f : M \to P$ is a surjective homomorphism of R-modules, there is an R-module homomorphism $g : P \to M$ such that $f \circ g$ is the identity map on P.

(a) Show that an R-module P is finitely generated projective *if and only if* there is a finitely generated free module F and an R-module P_1 such that $F \simeq P \oplus P_1$. (For "$\Rightarrow$" you need to show both P is finitely generated and P is projective, using the given definition of projective.)

(b) Let $P^* = \mathrm{Hom}_R(P, R)$ be the set of R-module homomorphisms from P to R. We can make P^* into an R-module by defining $(r\phi)(p) = r(\phi(p))$ for all $r \in R, p \in P, \phi \in P^*$. Show that if P is a finitely generated projective module then P^* is projective.

5. Let L/K be a finite Galois extension of fields and let $\alpha \in L$. Suppose there is an automorphism $\sigma \in \mathrm{Gal}(L/K)$ such that $\sigma(\alpha) = \alpha + 1$.

(a) Show that K has positive characteristic.

(b) If $L = K(\alpha)$, show that σ has order p, where p is the characteristic of K.

(c) Suppose σ generates $\mathrm{Gal}(L/K)$. Show that $\alpha^p - \alpha \in K$.

6. Let A_4, the group of even permutations of 4 objects, act on $\mathbf{C}^4$ by permuting the entries of the vectors. For example, the 3-cycle $(1, 2, 4)$ maps the vector (a, b, c, d) to (d, a, c, b). This yields a representation $\rho : A_4 \to GL_4(\mathbf{C})$.

(a) (3 points) Show that ρ contains the trivial representation as a subrepresentation.

(b) (7 points) Show that ρ is the sum of two irreducible representations. (*Note:* A_4 contains the identity, eight 3-cycles, and three products of two disjoint 2-cycles.) (*Hint:* A character is irreducible if and only if its inner product with itself is 1.)